SIMATS ENGINEERING

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Assignment 1 | CO1: Mathematical Foundations and Machine Learning Principles

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Scenario: Healthcare Risk Prediction	**Course Code:** CSA47

Task 1 – Patient Data Representation and Normalization (5 Marks)

a) Raw features. Age, blood pressure, glucose and cholesterol are all measured in different units and ranges. A feature with larger numbers (cholesterol, for example) can dominate distance calculations and gradients, which biases learning and slows down convergence.

b) Feature matrix. Columns are age, blood pressure, glucose and cholesterol. Sample records:

X = [[25, 120, 90, 180], [50, 140, 130, 220], [75, 160, 170, 260]]

Min-max normalization is applied column by column: x' = (x − min) / (max − min)

X' = [[0.0, 0.0, 0.0, 0.0], [0.5, 0.5, 0.5, 0.5], [1.0, 1.0, 1.0, 1.0]]

Check: Patient 2 age = (50 − 25)/(75 − 25) = 0.5 and glucose = (130 − 90)/(170 − 90) = 0.5.

c) Importance. Normalization stops large-scale features from dominating distance-based methods such as KNN and K-means. It also puts inputs on a comparable scale for gradient-based methods like neural networks, which usually improves numerical stability and convergence.

Task 2 – Vector Representation and Matrix Operations (2 Marks)

a) Let x = [age, glucose, blood pressure, cholesterol] and $w = [w_1, w_2, w_3, w_4]$. The weighted risk score is the dot product:

$$Risk\ score = w^T x = w_1(age) + w_2(glucose) + w_3(BP) + w_4(cholesterol)$$

b) Example. x = [50, 130, 140, 220] and w = [0.1, 0.3, 0.2, 0.05] *score = 0.1(50) + 0.3(130) + 0.2(140) + 0.05(220) = 5 + 39 + 28 + 11 =* ***83***

This is only a preliminary weighted score. It is not a clinically validated probability or a diagnosis.

Task 3 – Probability and Conditional Probability (6 Marks)

Let D = has disease and G = elevated glucose. Total = 1000, diseased = 200, non-diseased = 800, diseased with elevated glucose = 160, non-diseased with elevated glucose = 120.

P(D) = 200/1000 = 0.20 P(not D) = 0.80 P(G|D) = 160/200 = 0.80 P(G|not D) = 120/800 = 0.15

a) P(G) = (160 + 120)/1000 = 280/1000 = 0.28 = **28%**

b) P(D|G) = P(D and G)/P(G) = 160/280 = 0.5714 = **57.14%**

c) P(G|D) = 160/200 = 0.80 = **80%**

d) Conditional probability shows how disease status relates to a finding. About 57% of patientswith elevated glucose have the disease, so this can help prioritize further testing. It is not a diagnosis; other risk factors and confirmatory tests are still needed.

Task 4 – Bayesian Statistics for Disease Classification (4 Marks) Given: P(D) = 0.10, P(+|D) = 0.90, P(+|not D) = 0.05, P(not D) = 0.90.

$$P(+) = P(+|D)P(D) + P(+|\text{not } D)P(\text{not } D) = (0.90 \times 0.10) + (0.05 \times 0.90) = 0.135$$

$$P(D|+) = (0.90 \times 0.10) / 0.135 = 0.6667$$

a) The probability of disease given a positive test is **66.67%**.

b) The prior is the disease prevalence before testing. The likelihood is the probability of theobserved test result given the disease status. Bayes' theorem combines the two with the evidence to give the posterior.

c) Bayesian reasoning updates risk as new evidence arrives and accounts for prevalence andtest uncertainty. A positive result does not automatically mean the patient has the disease.

Task 5 – Entropy and Information Gain (3 Marks) *p(high) = 0.4, p(low) = 0.6*

$$H = -[0.4 \log_2(0.4) + 0.6 \log_2(0.6)] = -[0.4(-1.3219) + 0.6(-0.7370)] \approx 0.971 \text{ bits}$$

a) Entropy ≈ 0.971 bits. **b)** Entropy measures uncertainty or impurity; zero means every record is in one class, and for two classes a value near 1 bit means high uncertainty. **c)** Information gain = parent entropy − weighted child entropy after a split. A glucose split is useful if it lowers the total entropy of the resulting groups.

Task 6 – Numerical Computation and Distance Measurement (3 Marks)

A = (0.70, 0.80, 0.60) and B = (0.50, 0.75, 0.90)

$$d = \sqrt{[(0.70 - 0.50)^2 + (0.80 - 0.75)^2 + (0.60 - 0.90)^2]} = \sqrt{(0.0400 + 0.0025 + 0.0900)} =$$

$$\sqrt{0.1325} \approx 0.364$$

a) Euclidean distance = 0.364. **b)** A smaller distance means the patients have more similar feature values; a larger distance means they differ more. **c)** Normalization makes glucose, blood pressure and cholesterol comparable, so no feature dominates just because of its scale.

Task 7 – Bias–Variance Trade-off (4 Marks)

a) Model A has high bias (high training and validation error). Model B has high variance (verylow training error but high validation error). Model C generalizes well (low and similar errors).

b) A simple model may miss important patterns, which causes bias. Increasing complexityreduces bias but may increase variance because the model starts fitting noise.

c) Model A can be improved by increasing model capacity, adding useful features or reducingexcessive regularization. Model B can be improved with regularization, dropout, early stopping, more representative data or a simpler model.

Task 8 – Overfitting, Underfitting and Model Capacity (4 Marks)

a) A simple model underfits; a moderately complex model gives appropriate generalization; ahighly complex model with excellent training results but poor performance on unseen data is overfitting.

b) Model capacity is the ability of a model to represent relationships in data. Low capacitymisses complex patterns, while higher capacity can learn more complex relationships.

c) Increasing capacity first improves performance by reducing underfitting. If capacitybecomes excessive, the model memorizes noise and training examples, so generalization drops.

Validation and regularization help in choosing a suitable capacity.

Task 9 – Supervised Learning and Model Evaluation (3 Marks)

a) This is supervised learning because the historical records contain input features and knowndisease / disease-free labels. The model learns to predict labels for new patients.

b) Training data is used to learn the model parameters. A separate testing set estimatesperformance on unseen records and should not be used repeatedly for tuning.

Accuracy = (correct predictions / total predictions) × 100 = (92/100) × 100 = ***92%***

c) The model classified 92 of 100 test patients correctly. Sensitivity, specificity, precision,F1-score and the confusion matrix should also be checked, especially when the classes are imbalanced.

Task 10 – Unsupervised Learning and Patient Grouping (4 Marks)

a) Unsupervised learning is suitable because patient features are available but there are nopredefined risk labels; the aim is to discover patterns.

b) K-means steps: choose K centres; assign each patient to the nearest centre; recompute each centre as the mean of its assigned patients; repeat until the assignments or centres stop changing. Features should be normalized first.

c) K-means normally uses Euclidean distance, so patients with smaller distances across thechosen features end up in the same group.

d) Clusters can reveal patients with similar risk-factor profiles and guide further investigation. They are exploratory groups and must not replace medical diagnosis or clinician judgement.

Task 11 – Comparison of Supervised and Unsupervised Learning (4 Marks)

a) Dataset A has disease labels, so supervised learning is appropriate. Dataset B has no labels,so unsupervised learning can be used to find groups and patterns.

b) Supervised learning learns a mapping from features to known targets and outputspredictions, classes or probabilities. Unsupervised learning finds structure and outputs clusters or patterns, not verified disease labels.

c) A supervised classifier trained and validated on representative labelled records should beused to predict disease status for a new patient. Clustering can separately explore the unlabelled records, but cluster membership is not a diagnosis.

Task 12 – Design Challenge: Selecting and Adapting a Learning Strategy (8 Marks)

A. Impact Analysis (2 marks)

a) Going from 10,000 to 1,000,000 records increases matrix storage, preprocessing time andtraining computation. Efficient numerical libraries, batching, suitable hardware and scalable algorithms are needed. More data can improve learning if it is representative and reliable.

b) Normalize features to comparable scales, investigate missing values and use justifiedimputation or removal, and check variability and representativeness of the data to reduce bias.

B. Evaluation of Original Approach (2 marks)

a) A single supervised classifier is not enough for both objectives. It predicts disease statusfrom labelled examples but does not directly discover groups in unlabelled data.

b) Ignoring unlabelled records may miss useful patient subgroups. These records can beexplored with clustering or with carefully validated semi-supervised methods.

C. Adaptation or Redesign (2 marks)

a) Clean and normalize the data; train a supervised classifier on the labelled records fordisease-risk prediction; apply K-means (or another clustering method) on the normalized features to find patient groups, including patterns in unlabelled records; and have clinical experts review the profiles.

b) Use calibrated probability outputs for risk, distance measures on normalized features forclustering, and validation / regularization to choose model capacity and control overfitting. Large datasets are processed in batches or with scalable implementations.

D. Evidence-Based Defence (2 marks)

a) Compare the original and redesigned approaches on the same held-out test set. Forclassification, measure accuracy, precision, recall (sensitivity), specificity, F1, ROC-AUC, calibration and the confusion matrix. For clustering, assess silhouette score, cluster sizes, stability and clinical interpretability. Record runtime and memory use, and test alternative preprocessing steps.

b) Keep supervised learning for prediction and add unsupervised clustering for discovery.Normalization improves distances and optimization, probabilities express uncertainty, capacity control improves generalization, and scalable computation handles one million records.

Clinical validation and privacy safeguards are required before deployment.

Conclusion

Reliable healthcare prediction depends on sound feature representation, probability, numerical computation and the right learning methods. A combined supervised and unsupervised workflow can predict disease risk from labelled records and explore patterns in unlabelled

records. Normalization, missing-value handling, validation and clinical review all improve reliability.